

Native Platforms

Native **SwiftUI (MVVM)** on iOS (live on Apple Testflight) and **Kotlin Compose** on Android. Operates offline-first; all lessons execute with zero network connectivity.

Stage Grammar

6 modules and 36 lessons: bundled as JSON. 4 interactive task types: Quiz, Spot-It, Sort-It, Build-It. Mistakes are never punished (explanations shown after 2 misses).

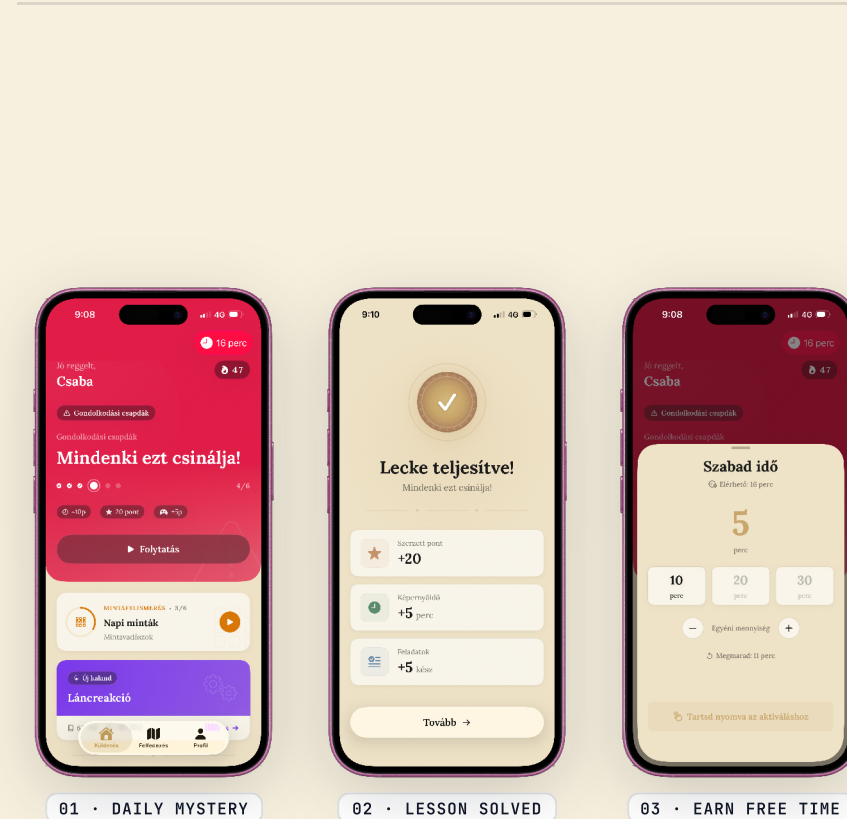
Premack Contingency

Applies the Premack [1959] differential probability law. Solving lesson credits the ledger with **+20 pts & +5 min**. Child spends balance within parent-allowed hours.

Constrained Gemini

System prompt restricts Gemini to Socratic dialogue: probes reasoning and gives hints, never direct answers. Strict low token cap prevents lecturing; unlocks only post-task.

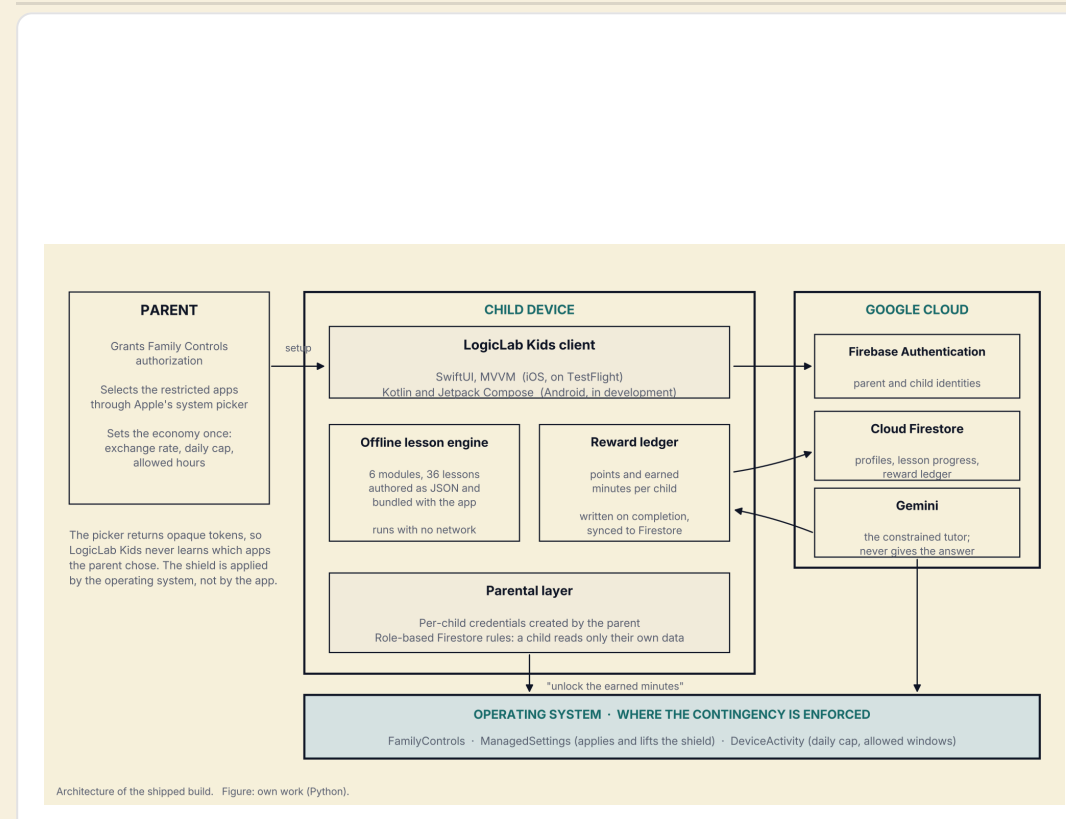
Interactive Child Experience • iOS & Android



The Behavioral Lock

Screen entertainment is the highest-probability behavior; structured reasoning practice is among the lowest. The lock and key are in the OS: a child cannot negotiate around it.

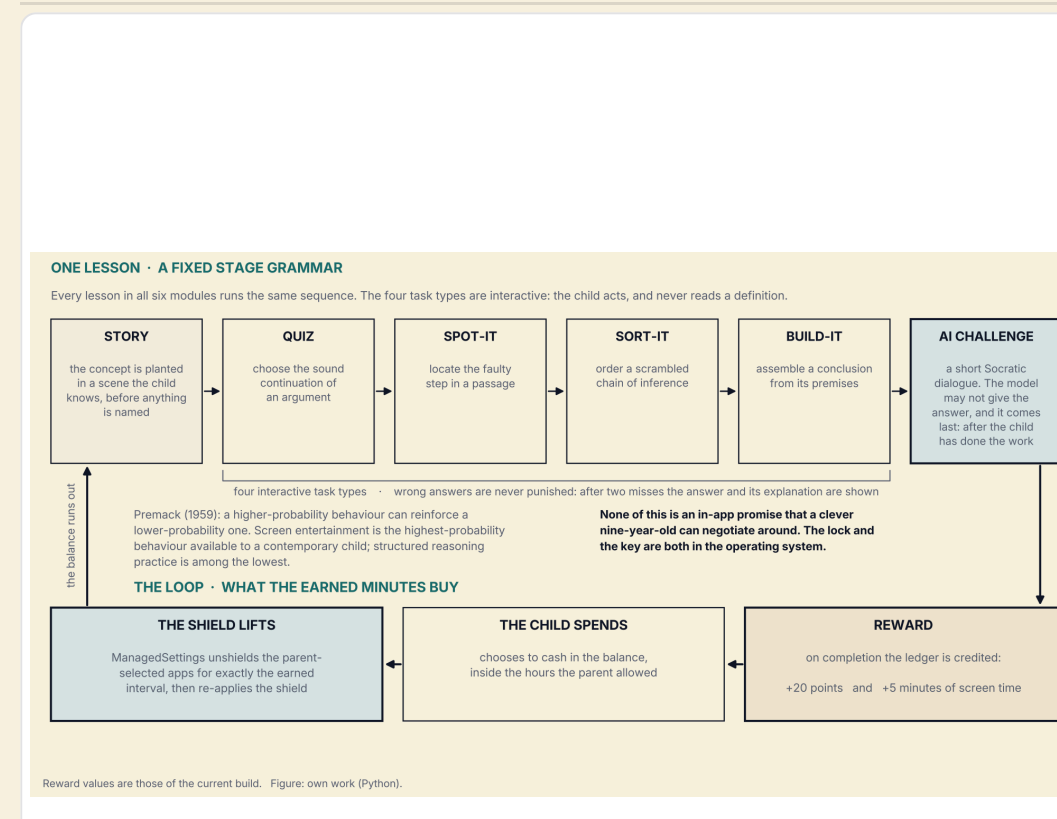
System Architecture



Opaque Token Shielding

The system picker returns **opaque tokens**, so LogicLab Kids never access selected. Shields are applied directly by iOS ManagedSettings, not in-app.

Stage Grammar & Loop



4 Core Task Type

Quiz (sound continuation), **Spot-It** (faulty step), **Sort-It** (inference chain), and **Build-It** (conclusion assembly). The child acts dynamically rather than memorizing definitions.

REFERENCES